1 The Group 17 elements are oxidising agents.

(a) **(i)** Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

(d) Table 1.3 gives some information about $MgCl_2$ and $SiCl_4$.

Table 1.3

	$MgCl_2$	**$SiCl_4$**
electrical conductivity when liquid	conducts	does **not** conduct
observation when added to water	dissolves	vigorous reaction

(i) Explain the difference between the electrical conductivity of liquid $MgCl_2$ and of liquid $SiCl_4$. Refer to bonding and relevant particles in your answer.

.....

.....

.....

.....

..... [2]

2 Aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, are both used as reagents and catalysts.

(a) The melting point of Al_2O_3 is 2072 °C. The melting point of P_4O_{10} is 340 °C.

Explain the difference in the melting points of these two compounds.

.....

.....

.....

..... [3]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(c) Acidified dichromate(VI) ions will convert methanal, CH_2O, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

(iv) Methanal is a liquid at –30 °C but carbon dioxide is a gas at this temperature. Explain why.

.....

.....

.....

..... [2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) **(i)** Explain why the elements Na to A*l* are good electrical conductors.

.....

..... [1]

(ii) Explain why the elements P, S and C*l* do not conduct electricity.

.....

..... [1]

(b) Fig. 2.1 shows the variation in melting point of the Period 3 elements Si to C*l*.

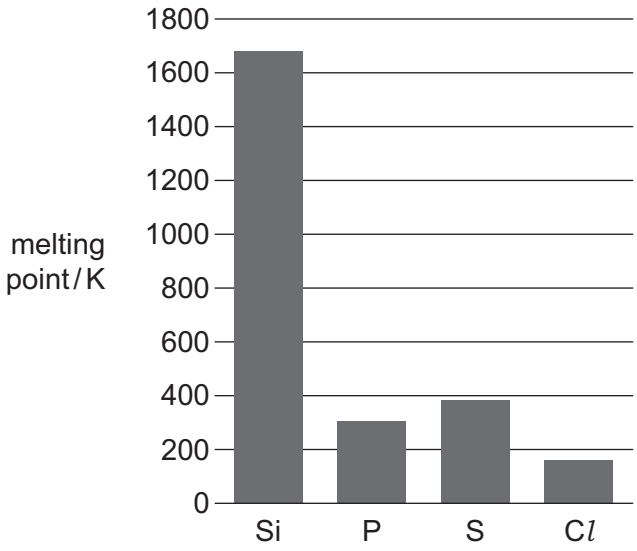


Fig. 2.1

The Period 3 elements Si to C*l* are all non-metals.

Explain why there is a large difference between the melting point of Si and the melting points of P, S and C*l*.

.....

.....

.....

..... [2]

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

.....

..... [2]

(d) Hexane, C_6H_{14}, has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

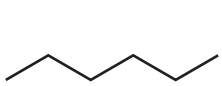
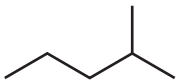
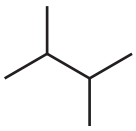
hexane	A	B	C	D
				

Fig. 3.1

(i) Complete Fig. 3.1 by drawing structures for **C** and **D**, the other two structural isomers of hexane. [2]

(ii) **A**, **B** and hexane have different boiling points.

Arrange **A**, **B** and hexane in order of increasing boiling point.

Explain your answer.

lowest ..... < ..... < ..... highest

.....

.....

.....
[3]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(c) Acidified dichromate(VI) ions will convert methanal, CH_2O, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

(iv) Methanal is a liquid at –30 °C but carbon dioxide is a gas at this temperature. Explain why.

.....

.....

.....

..... [2]

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(a) Complete Table 1.1 by identifying the lattice structures of the crystalline solids of Mg, Si and P.

Table 1.1

element	Mg	Si	P
lattice structure in crystalline solid			

[3]

(b) The relative electrical conductivities of the Period 3 elements are shown in Fig. 1.1.

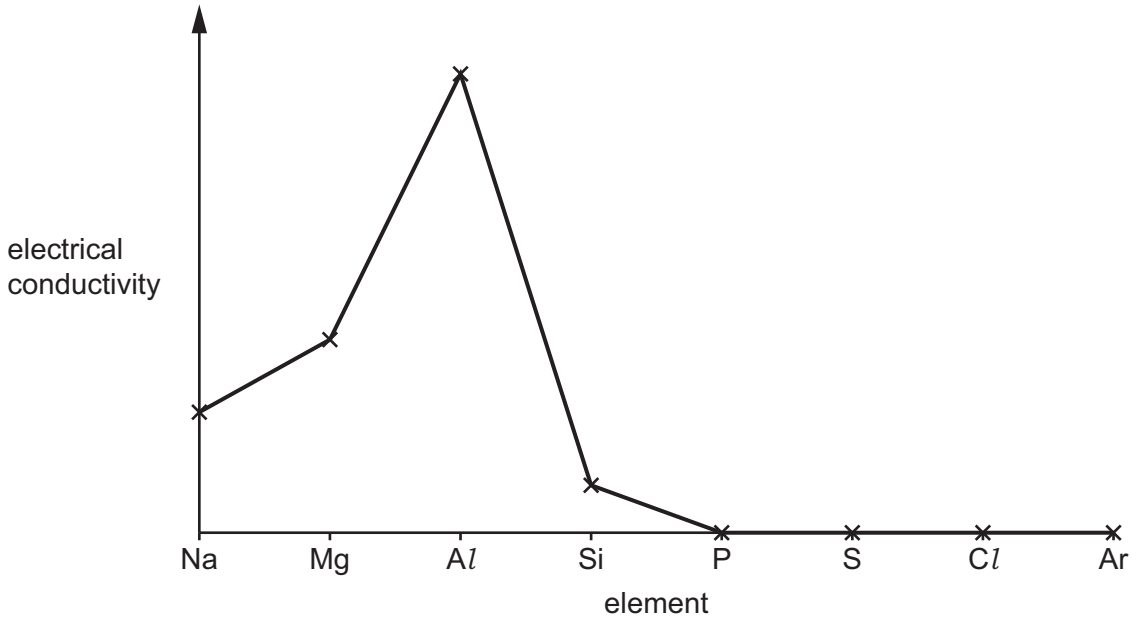


Fig. 1.1

Explain why there is an increase in conductivity from Na to Al and why P, S and Cl are non-conductors of electricity.

.....

.....

.....

.....

..... [2]

6 **P**, **Q** and **R** are three different hydrocarbon molecules.

(a) **(i)** Define empirical formula.

.....

..... [1]

(b) **Q** is a volatile hydrocarbon.
0.194 g of gaseous **Q** occupies a volume of 71.5 cm^3 at 100 °C and 100 kPa.

Use the ideal gas equation to calculate the M_r of **Q**.

M_r of **Q** = ... [2]

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water. State the formula of the oxide.

..... [1]

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point / °C
SO_2	–73
H_2O	0
SO_3	17
SiO_2	1610
Na_2O	1132
MgO	2852
Al_2O_3	2072

(i) Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

..... [1]

(ii) A student suggests the following hypothesis.

The melting point of an ionically bonded oxide is only determined by the charge on the cation.

Use Table 2.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

.....

..... [2]

1 **(a)** Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.

(i) Name the type of lattice structure present in the crystalline solids diamond and graphite.

..... [1]

(ii) Explain how graphite conducts electricity.

.....

..... [1]

(iii) Explain why diamond does **not** conduct electricity.

.....

..... [1]

(b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

(i) Write an equation for each reaction.

.....

..... [2]

(ii) Describe the appearance of phosphorus(V) chloride and silicon(IV) chloride at room temperature.

phosphorus(V) chloride .....

silicon(IV) chloride .....
[2]

(d) The melting points of different oxides are shown in Table 1.1.

Table 1.1

oxide	melting point / °C	force of attraction broken during melting
SO_2	–73	
H_2O	0	
SiO_2	1610	
Na_2O	1132	
MgO	2852	

(i) Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

i.d. = instantaneous dipole–induced dipole

p.d. = permanent dipole–permanent dipole

H = hydrogen bond

C = covalent bond

I = ionic bond

[3]

(ii) A student suggests the following hypothesis.

The stronger the covalent bond between atoms in non-metal oxides, the higher the melting point.

Use Table 1.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

.....

..... [2]

1 **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

(i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium .....

bonding in sodium oxide .....
[1]

(ii) Explain how solid sodium conducts electricity.

.....

..... [1]

(d) Table 1.2 shows melting points of some oxides.

Table 1.2

oxide	melting point / °C
SO_2	–73
H_2O	0
SO_3	17
SiO_2	1610
MgO	2852
Al_2O_3	2072

A student suggests the following hypothesis.

The higher the oxidation number of the element combined with oxygen, the higher the melting point of the oxide.

Use Table 1.2 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

.....

..... [2]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(iii) The third compound of phosphorus and chlorine, **W**, has a relative molecular mass, M_r, between that of PCl_3 and PCl_5. The compound contains 69.6% by mass of chlorine.

Determine the molecular formula of **W**.

molecular formula of **W** = ... [2]

(iv) **W** is a liquid at room temperature and pressure. It reacts vigorously with water to form an acidic solution.

Suggest the structure and bonding in **W**. Explain your answer.

.....

.....

..... [2]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(ii) Identify the lattice structure shown by graphite.

..... [1]

(iii) Explain why Sn has good electrical conductivity.

.....

..... [1]

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

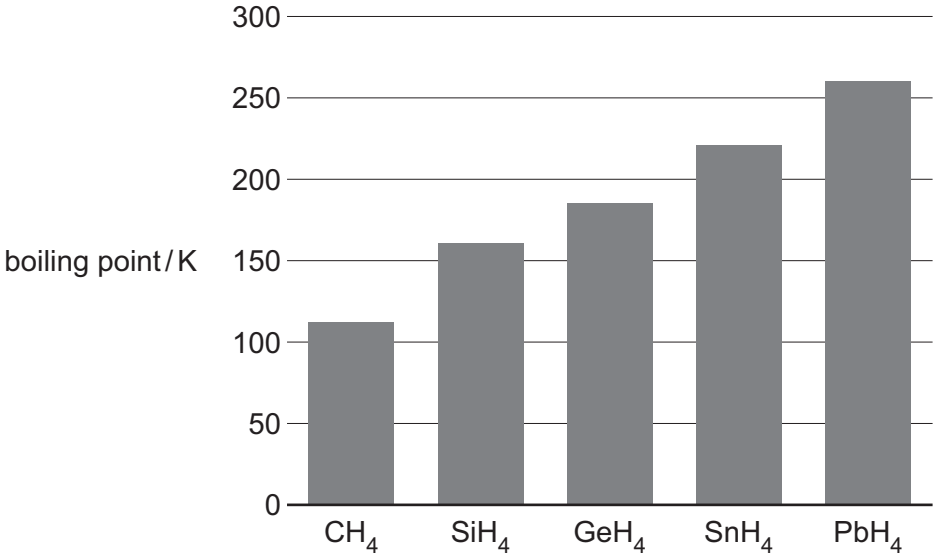


Fig. 3.2

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

.....

.....

.....

..... [2]

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

..... [1]

(iii) SiO_2 is a white solid that melts above 1700 °C.

$SiCl_4$ is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

.....

.....

.....

.....

..... [2]

2 Oxygen is a Group 16 element.

(a) **(i)** Write equations for the following reactions.

- sodium and oxygen

.....

- sulfur and oxygen

.....

[2]

(c) Fig. 2.1 shows the boiling points of H_2O and other Group 16 hydrides.

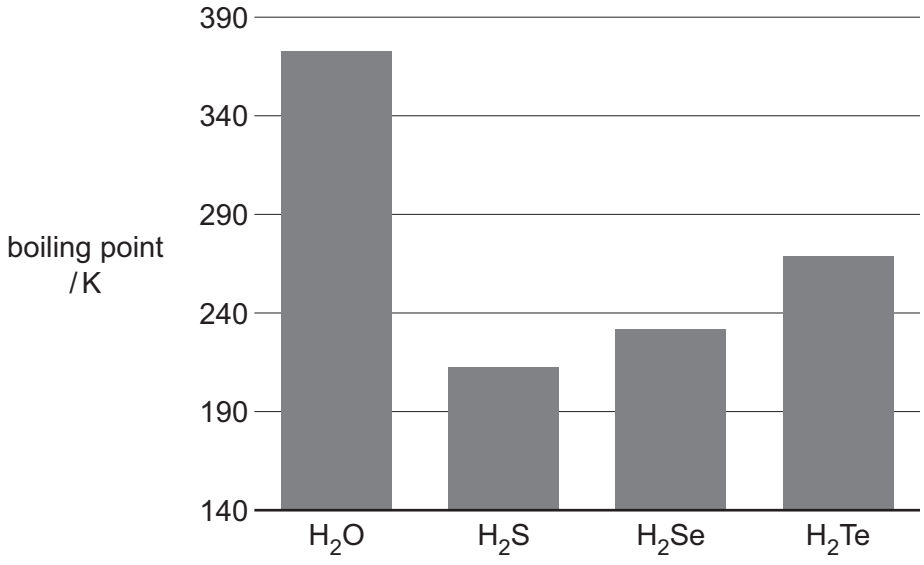


Fig. 2.1

(i) Explain the trend in the boiling points of the Group 16 hydrides H_2S to H_2Te.

.....

.....

.....

..... [2]

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(i) Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....

[2]

(d) Complete Table 3.1 to show the properties of nitrogen and phosphorus in their standard states.

Table 3.1

	nitrogen	phosphorus
state and appearance of standard state	colourless gas	white solid
electrical conductivity		poor
type of bonding		
type of structure	simple	

[2]

(e) A form of solid nitrogen has a lattice structure similar to solid iodine.

Identify the type of lattice structure of solid nitrogen.

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(ii) Identify the lattice structure shown by graphite.

..... [1]

(iii) Explain why Sn has good electrical conductivity.

.....

..... [1]

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

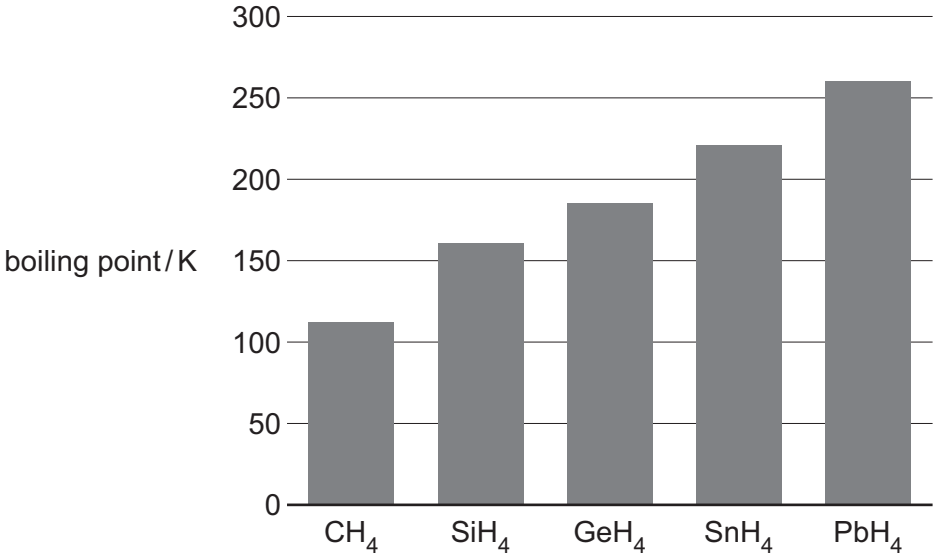


Fig. 3.2

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

.....

.....

.....

..... [2]

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

..... [1]

(iii) SiO_2 is a white solid that melts above 1700 °C.

$SiCl_4$ is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

.....

.....

.....

.....

..... [2]

3 **(a)** Write an equation to show the reaction for the standard enthalpy change of formation of H_2O. Include state symbols.

..... [2]

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000 °C.

(i) Suggest the structure and bonding in boron carbide.

..... [1]

1 **(a)** Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

(c) Draw a labelled diagram to show the structure and bonding in sodium.

[1]

(d) Fig. 1.1 shows the variation in melting point of some Period 3 elements in their standard states at room temperature and pressure.

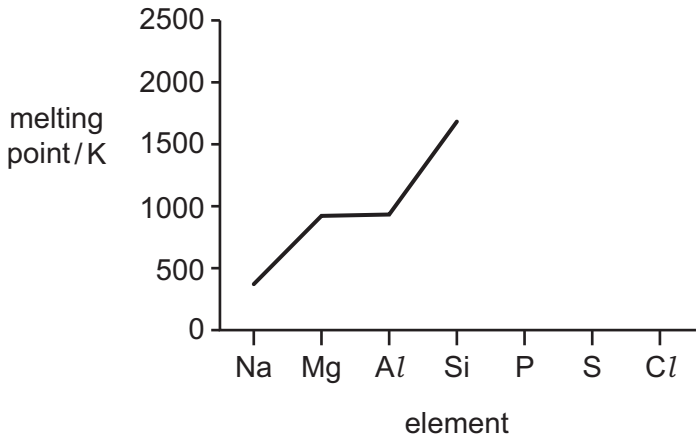


Fig. 1.1

(i) Explain why Si has a high melting point.

.....

.....

..... [1]

2 **(a)** Sulfur chloride, SCl_2, is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

..... [1]

(d) Boron nitride is a white solid that melts above 2900 °C.

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

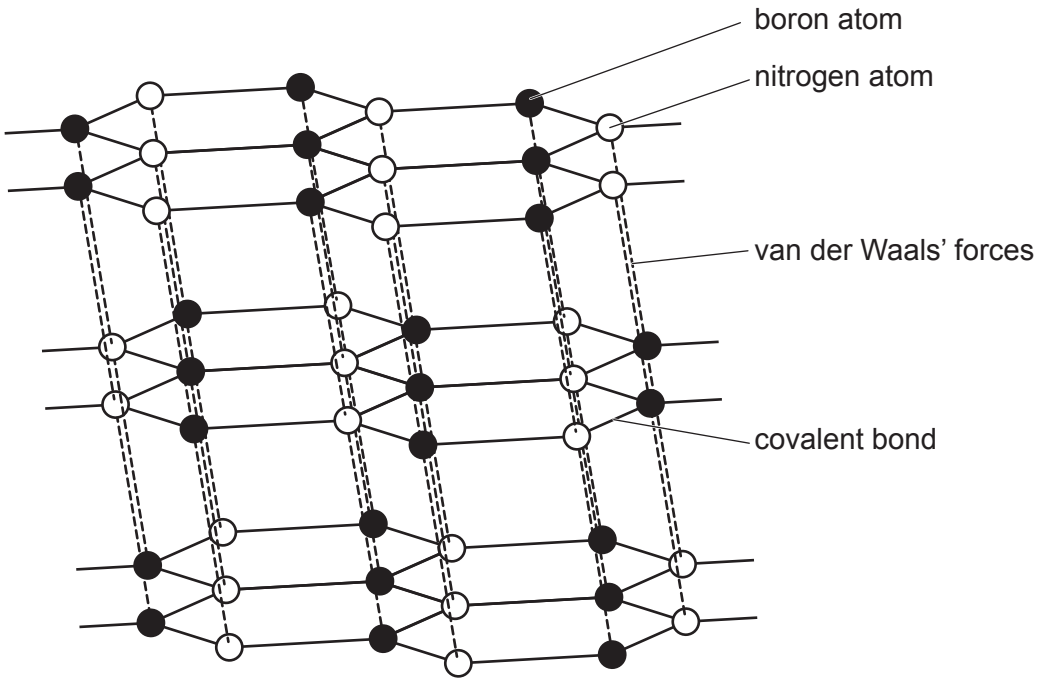


Fig. 2.2

(i) Use Fig. 2.2 to deduce the empirical formula of boron nitride.

..... [1]

(ii) Suggest the identity of another crystalline solid that has atoms arranged in layers similar to that of solid boron nitride.

..... [1]

1 Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

(b) Bismuth reduces water to form bismuth oxide, Bi_2O_3. A colourless gas that ignites with a squeaky pop also forms.

(i) Construct an equation for the reduction of water by bismuth.

..... [1]

(ii) Bi_2O_3 is a yellow insoluble solid that melts at 1090 K. The molten compound conducts electricity.

Deduce the structure and bonding of Bi_2O_3. Explain your answer.

.....

.....

..... [2]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(d) $POCl_3$ shows similar chemical properties to PCl_5.

$POCl_3$ has a melting point of 1°C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(d) Molten $KClO_3$ reacts with glucose, $C_6H_{12}O_6$.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

$KClO_3$ melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

(i) Use the ideal gas equation to calculate the volume, in m^3, of one mole of gas at 630 K and 1.00×10^5 Pa.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = ... m^3
[1]

(ii) 5.00 g of $C_6H_{12}O_6$ reacts completely with molten $KClO_3$.

Use your answer to **(d)(i)** to calculate the total volume of gas released at 630 K and 1.00×10^5 Pa in this reaction.

(If you were unable to answer **(d)(i)**, use 0.0463 m^3 in this question. This is **not** the correct answer to **(d)(i)**.)

total volume of gas released = ... m^3
[2]

1 The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

(a) (i) Describe the variation in atomic radius from silicon to sulfur.

..... [1]

(ii) The melting point of silicon is 1410 °C. The melting point of sulfur is 113 °C.

Explain this difference.

.....

.....

.....

..... [3]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(d) $POCl_3$ shows similar chemical properties to PCl_5.

$POCl_3$ has a melting point of 1°C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(d) Molten $KClO_3$ reacts with glucose, $C_6H_{12}O_6$.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

$KClO_3$ melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

(i) Use the ideal gas equation to calculate the volume, in m^3, of one mole of gas at 630 K and 1.00×10^5 Pa.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = ... m^3
[1]

(ii) 5.00 g of $C_6H_{12}O_6$ reacts completely with molten $KClO_3$.

Use your answer to **(d)(i)** to calculate the total volume of gas released at 630 K and 1.00×10^5 Pa in this reaction.

(If you were unable to answer **(d)(i)**, use 0.0463 m^3 in this question. This is **not** the correct answer to **(d)(i)**.)

total volume of gas released = ... m^3
[2]

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(a) (i) Identify the lattice structure of copper.

..... [1]

1 The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

(a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium .....

bonding in sodium chloride .....
[1]

(ii) Explain the difference in the melting points of magnesium and sodium chloride.

.....

..... [1]

(iii) Name the strongest intermolecular force that exists between NH_3(l) molecules.

..... [1]

(v) The melting points of ice and ammonia are shown in Table 1.2.

Table 1.2

solid	melting point / K
ice	273
ammonia	195

Suggest **two** reasons for the difference in the melting points of ice and ammonia.

.....

.....

..... [2]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy / kJ mol^{-1}	—	7300	2350	3390	2260

(a) (i) Define electronegativity.

.....

..... [1]

(e) (i) Complete Table 1.2 by placing a tick (✓) to show which of the compounds have molecules with an overall dipole moment.

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

[2]

(ii) At 150 °C and 103 kPa, all of the compounds listed in Table 1.2 are gases.

Under these conditions, 0.284 g of one of the compounds occupies a volume of 127 cm^3.

Use this information to calculate the M_r of the compound. Hence, identify the compound from those given in Table 1.2.

Show your working.

M_r = ... identity of compound = ...

[3]

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(a) **(i)** Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

(ii) Explain why Mg has a higher electrical conductivity than Na.

.....

..... [1]